LABSHEET-3

1.Create an HTML page using JavaScript to accept two numbers and perform addition, subtraction, multiplication, and division.

```
<!DOCTYPE html>
<html>
<head>
  <title>Arithmetic Operations</title>
</head>
<body>

  <h2>Arithmetic Operations</h2>

  <label>Enter First Number:</label>
  <input type="number" id="num1"><br><br>

  <label>Enter Second Number:</label>
  <input type="number" id="num2"><br><br>

  <button onclick="calculate()">Calculate</button>

  <h3>Results:</h3>
  <p id="result"></p>

  <script>
    function calculate() {
      let a = parseFloat(document.getElementById("num1").value);
```

```
        let b = parseFloat(document.getElementById("num2").value);

        if (isNaN(a) || isNaN(b)) {
          document.getElementById("result").innerHTML =
            "Please enter both numbers.";
          return;
        }

        let addition = a + b;
        let subtraction = a - b;
        let multiplication = a * b;
        let division = b !== 0 ? a / b : "Cannot divide by zero";

        document.getElementById("result").innerHTML =
          "Addition: " + addition + "<br>" +
          "Subtraction: " + subtraction + "<br>" +
          "Multiplication: " + multiplication + "<br>" +
          "Division: " + division;
      }
    </script>

</body>
</html>
```

2. Write a JavaScript program to accept a student's marks and display the grade according to the following criteria:

- 90–100: A+
- 80–89: A
- 70–79: B
- 60–69: C
- Below 60: Fail

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Grade Calculator</title>
</head>
<body>

  <h2>Student Grade Calculator</h2>

  <label>Enter Student Marks (0-100):</label>
  <input type="number" id="marks">
  <br><br>

  <button onclick="checkGrade()">Check Grade</button>

  <h3 id="result"></h3>

  <script>
    function checkGrade() {
      let marks = Number(document.getElementById("marks").value);
      let grade;
```

```
    if (document.getElementById("marks").value === "") {
        grade = "Please enter marks.";
    }
    else if (marks < 0 || marks > 100) {
        grade = "Please enter marks between 0 and 100.";
    }
    else if (marks >= 90) {
        grade = "A+";
    }
    else if (marks >= 80) {
        grade = "A";
    }
    else if (marks >= 70) {
        grade = "B";
    }
    else if (marks >= 60) {
        grade = "C";
    }
    else {
        grade = "Fail";
    }

    document.getElementById("result").innerHTML =
        "Grade: " + grade;
}
```

```
  </script>


</body>

</html>
```

3.Write a JavaScript program to accept a number from the user and display its multiplication table using a loop.

```
<!DOCTYPE html>

<html>

<head>

  <title>Multiplication Table</title>

</head>

<body>


  <h2>Multiplication Table</h2>


  <label>Enter a Number:</label>

  <input type="number" id="num">

  <br><br>


  <button onclick="showTable()">Show Table</button>


  <h3>Result:</h3>

  <p id="result"></p>


  <script>

    function showTable() {
```

```
        let num = document.getElementById("num").value;
        let result = "";

        if (num === "") {
           document.getElementById("result").innerHTML =
              "Please enter a number.";
           return;
        }

        num = Number(num);

        for (let i = 1; i <= 10; i++) {
           result += num + " x " + i + " = " +
                 (num * i) + "<br>";
        }

        document.getElementById("result").innerHTML = result;
     }
  </script>

</body>
</html>
```

4. Create a JavaScript function to calculate the factorial of a number entered by the user.

```
<!DOCTYPE html>
<html>
<head>
```

```
  <title>Factorial Calculator</title>
</head>
<body>

  <h2>Factorial Calculator</h2>

  <label>Enter a Number:</label>
  <input type="number" id="num">
  <br><br>

  <button onclick="calculateFactorial()">Calculate Factorial</button>

  <h3 id="result"></h3>

  <script>
    function calculateFactorial() {
      let num = Number(document.getElementById("num").value);
      let factorial = 1;

      if (document.getElementById("num").value === "") {
        document.getElementById("result").innerHTML =
          "Please enter a number.";
        return;
      }

      if (num < 0 || !Number.isInteger(num)) {
```

```
        document.getElementById("result").innerHTML =
          "Please enter a non-negative integer.";
        return;
      }

      for (let i = 1; i <= num; i++) {
        factorial = factorial * i;
      }

      document.getElementById("result").innerHTML =
        "Factorial of " + num + " is " + factorial;
    }
  </script>

</body>
</html>
```

5. Create a simple Student Registration Form and use JavaScript to validate Name, Email, Mobile Number, and Password before submitting the form.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Registration Form</title>
</head>
<body>

  <h2>Student Registration Form</h2>
```

```
<form onsubmit="return validateForm()">

  <label>Name:</label>
  <input type="text" id="name">
  <br><br>

  <label>Email:</label>
  <input type="text" id="email">
  <br><br>

  <label>Mobile Number:</label>
  <input type="text" id="mobile">
  <br><br>

  <label>Password:</label>
  <input type="password" id="password">
  <br><br>

  <button type="submit">Register</button>

</form>

<p id="result"></p>

<script>
```

```
function validateForm() {
  let name = document.getElementById("name").value.trim();
  let email = document.getElementById("email").value.trim();
  let mobile = document.getElementById("mobile").value.trim();
  let password = document.getElementById("password").value;

  let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  let mobilePattern = /^[0-9]{10}$/;

  if (name === "") {
    alert("Please enter your name.");
    return false;
  }

  if (!emailPattern.test(email)) {
    alert("Please enter a valid email address.");
    return false;
  }

  if (!mobilePattern.test(mobile)) {
    alert("Please enter a valid 10-digit mobile number.");
    return false;
  }

  if (password.length < 8) {
    alert("Password must contain at least 8 characters.");
```

```
            return false;
        }

        document.getElementById("result").innerHTML =
            "Registration Successful!";

        return false;
    }
  </script>

</body>
</html>
```